

PROBLEM STATEMENT, HOW SEARCH WORKS AND RETRIEVAL JOURNEY

Problem statement : Users struggle to retrieve Google Photos memories they remember but can't precisely describe.

Goal: increase successful retrieval of a photo user remembers using **vague prompts** . **Not about improving search** broadly rather its about understanding memory, where retrieval breaks, and where to intervene.

Ask Photos

Ask Photos is Google's AI search. It understands your question, finds photos, and reads them, including any text inside the image to write an answer and select which photos to show.

◆ Powered by multimodal AI

Gemini reasons across text, images, and visual context together and not isolated keywords. It can interpret a photo's contents and text inside it, not just filename or metadata.

What it searches on


Who


When


Where


What's Visible

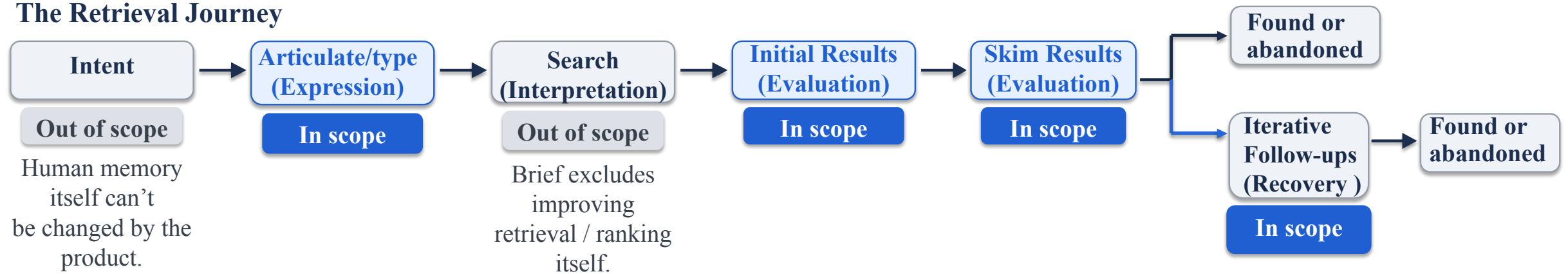

Text in Image


What You Want Back

 **Vague Prompt, Defined:** A prompt is vague when it lacks a precise anchor in at least 2 of 3 filters(Person, Time, Location) using approximations ("a few years ago," "that café") instead of exact facts.

Note: This is our working definition for this project — not an official definition used by Google Photos.

The Retrieval Journey



NOTE: Manual retrieval — scrolling the full timeline, browsing albums, or Memories with no search involved — is out of scope for this project. Scrolling *within* the search results shown above (Skim Results) remains in scope, as part of the search journey itself.

AI DISCOVERY ENGINE: SYSTEM OVERVIEW, WORKFLOW & OUTPUTS

1,616 Search Reviews Analyzed

Filtered from 5 platforms to only genuine search attempts (manual browsing excluded).

4 Stages Tracked

Finding the Words, Being Understood, Spotting the Right Photo, Getting Help When Stuck

100.0% (1,616 / 1,616) Verified

Every stat is traceable to a real, word-for-word user quote.

Fast, Live Search

Fast, Live Search: the review database is searched locally in seconds, with AI-generated answers.

What This Engine Is

The problem:

Star ratings tell you how happy users are, not why they couldn't find a photo — forgotten dates, cluttered libraries, blank search screens.

What this does:

Turns thousands of scattered complaints into clear, countable product opportunities, grounded in real user words.

Where the data comes from:

Google Play Store, Apple App Store, Reddit, Hacker News, and YouTube — filtered down to reviews specifically about searching for a photo, not general app complaints or manual scrolling.

How It Was Built

1 Gather: Automatically collect years of reviews and discussions across all 5 platforms.

2 Clean : Filter out irrelevant or duplicate reviews — including near-duplicate reviews reworded slightly, caught with a similarity check, not just exact copies.

3 Classify: An AI model (Gemini) reads each review and tags which part of the search journey it's about, and what specific detail the user forgot.

4 Search & Explore: Every review is indexed for instant lookup, so anyone can ask a question and get a real, quote-backed answer in seconds.

Built With

AI Understanding

Google Gemini reads and classifies each review's meaning.

Fast Local Search

A lightweight search index (no external server needed) returns matches almost instantly.

Reliable Data Storage

A structured local database keeps every review, tag, and quote organized and traceable.

Live Dashboard

A modern web app (Next.js) where anyone can browse findings or ask questions directly.

What It Delivers

 Exactly where retrieval breaks:
59.0% (953 / 1,616 reviews) — users struggle to find the right words to describe what they remember.

38.9% (628 / 1,616 reviews) — the search doesn't correctly understand what they described.

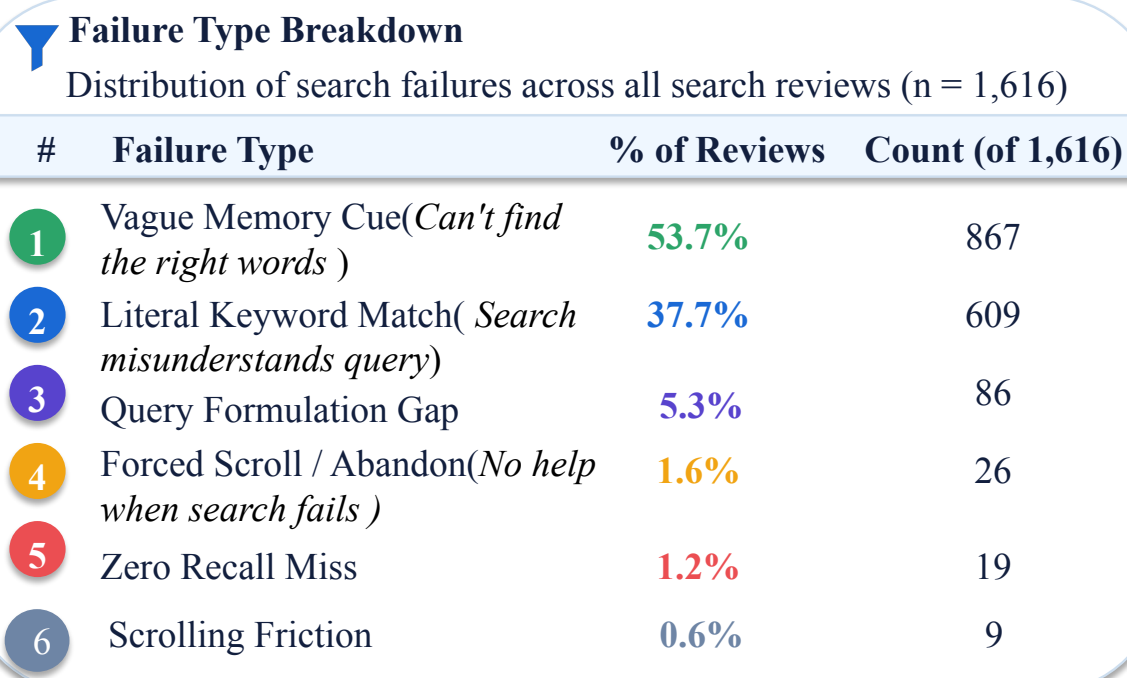
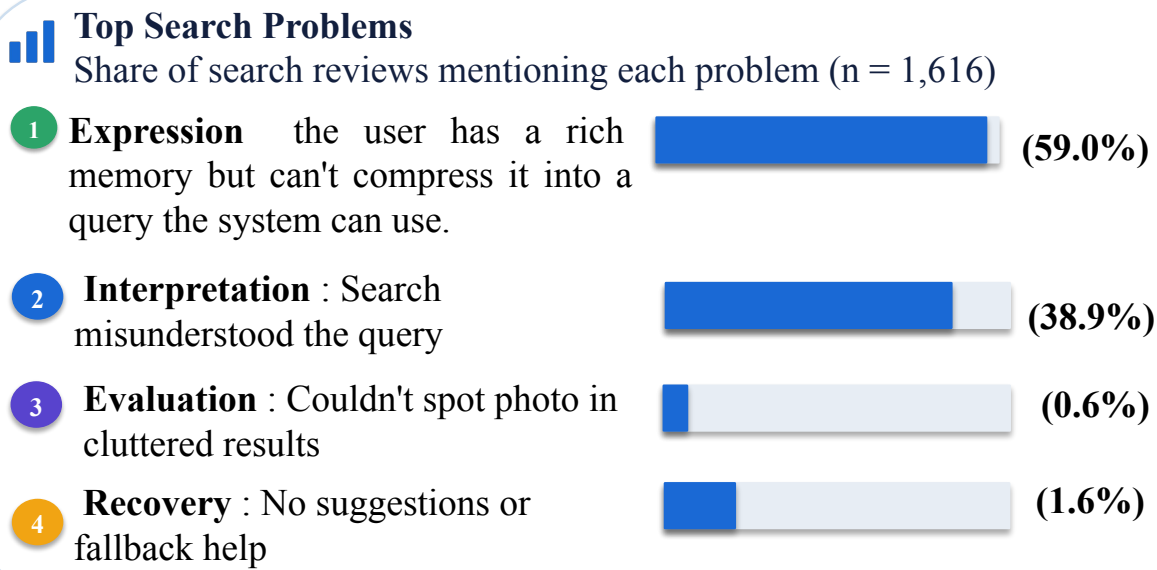
 10 deep-dive breakdowns: covering forgotten dates, mismatched faces/pets, screenshot clutter, and endless scrolling.

 Every claim backed by a real quote: **100.0% (1,616 / 1,616 reviews)** traceable to an exact user statement — nothing invented.

 Ask it yourself: A built-in AI assistant lets anyone ask a question in plain English and get a real, evidence-backed answer instantly.

EXPRESSION IS THE LARGEST PROBLEM WE CAN ACT ON, FOUND USING THE AI DISCOVERY ENGINE

AI Discovery Engine Findings



Business Outcome / North Star Metric
Vague Query Success Rate : % of search sessions with a **vague prompt** that end in **successful retrieval** of the photo (first try, after refinement, or via follow-up).

Defining successful retrieval: the person finds the photo they were looking for, within one search session. **Counted as found when they:** share it, edit it, add it to an album, or view it for 5+ seconds without searching again.

NOTE: this is a proxy. We can't see what's in someone's head, so we infer success from behavior. It's a working definition for this project, not an official Google Photos metric.

Product Outcomes

Query Formation Rate
Do people include more than one detail when searching from memory?
% of of vague-prompt sessions where the first query includes 2+ distinct cue type(person,time, location) and not just one weak keyword.

Refinement Attempt Rate
Do people try again after a search fails, instead of giving up?
% of failed searches where the user makes another attempt rather than abandoning to manual scrolling.

Retry Success Rate
When people try again after a failed search, do they find the photo?
% of retries, after a failed search, that end in successful retrieval.

PEOPLE REMEMBER THE MOMENT, BUT THE WORDS THEY TYPE MATCH MANY PHOTOS

Subtitle: The engine labels Expression "finding the right words"; our research shows people find words, but the words match too many photos.

Target segment

Google Photos users with Ask Photos(Gemini AI) turned on who search for a old photos from vague, multi-cue memory (person,time, location) and not the exact date.

Why this segment

- 1 **They remember the moment,** not the details: the occasion, who was there and roughly when
- 2 **They live the problem.**

Retrieval failure only happens at the moment someone searches from a vague memory. People who never search, or who remember exact dates, never hit it thus fixing their experience wouldn't move the metric.

- 3 **Their searches meet the Gemini layer**

With Ask Photos on, Google decides which experience answers, and complex queries go to Ask Photos.

Hypotheses

- H1 People remember the occasion and feeling of a photo, but those are hard to type
- H2 The words they type (an object, a person, a place) match many photos
- H3 People are guessing the words the system wants
- H4 With no date, people use loose time like “last summer” or “young”
- H5 For a known person, people skip typing and use the People tab

Survey findings

- 33 responses; 20 answered the search questions
- 1 10 of 20 recalled the occasion or feeling; the searches they described named objects, people and places (H1)
 - 2 16 of 20 recalled 2+ details, yet 12 of 20 typed a single word at some point (H2)
 - 3 Only 1 of 20 said finding words was their biggest problem; 12 of 20 said the app didn't understand them (H3)
 - 4 12 of 20 recalled roughly when it happened (H4)
 - 5 9 of 20 remembered who was there (H5)

User interview findings

5 interviews

- 1 Arun and Kaarthickeyan search by the most vivid object or scene (H1)
- 2 Kaarthickeyan's "swimming pool" search returned 83 results. Ithihas: a general prompt like "just me" gives "thousands of photos". A distinctive one worked: Arun's "silver racket". (H2)
- 3 Harshini: “I feel like my wording was correct only” (H3)
- 4 Ithihas used his age; Kaarthickeyan used “young” (H4)
- 5 Arun and Harshini use the People tab; Aditya (contrast case) searches by a tagged name (H5)

Hypothesis validation

H1 ✓ Supported	H2 ✓ Supported	H3 ✗ Not supported	H4 ● Partially supported	H5 ✓ Supported
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RETRIEVAL BREAKS AT THE TYPING STEP: PEOPLE REMEMBER THE SCENE, BUT THE WORDS THEY TYPE MATCH MANY PHOTOS

1 Target user segment and retrieval scenario

Users who search for a specific old photo from a vague memory (defined on the previous slide).

Retrieval scenario: They want one specific old photo. They remember the occasion, who was there and roughly when, but not the date or exact place. They type a few words, and the photo they know exists doesn't show up .



ALEX: the short-query searcher

Goal: Find one photo fast, without scrolling for years.

Needs: to have the few words he types point to that one photo

Job to be done: When I remember a photo but not its date, I want to type what I remember and reach it, so I don't end up scrolling .



PRIYANKA: the guesser

Goal: Find an old photo she can picture, without scrolling through years of photos .

Needs: to describe a photo in a way that singles it out

Job to be done: when I remember an old photo by how it looked, I want to describe it in a way that singles it out, so I don't keep guessing

5 Why solving it creates user value

A photo they know exists stops going unfound. Today 15 of 20 need a couple of tries, 16 of 20 see unrelated results, and 14 of 20 end up scrolling.

2. Product outcome we intend to influence

Query Formation Rate: the % of vague searches whose first query has 2+ cue types. It comes first, since Refinement and Retry only happen after a miss.

3 Root cause of retrieval failure

People remember a photo as a scene of several details, but the words they tend to type for it (a pool, "me", a beach) match many photos in a large library. So the query often fails to single out the one they mean.

Why: The details that would single it out (the occasion, the feeling, how it looked) are often not put into words. Only the combination of them pinpoints the photo, and the combination is what gets lost.

4 Existing user workarounds

Scroll to a date: 14 of 20 after a miss

Retry with new words: works quickly for 9 of 20, only sometimes or rarely for 9

Leave the app: 8 of 20 ask a friend or search WhatsApp

People tab: works only for tagged people

Each one goes around the query. None changes what it carries.

6 Why it makes business sense for Google Photos

Google earns from paid cloud storage that Photos, Drive and Gmail share. Photos is one of the products that fill that storage, so Google benefits when people keep trusting it with a growing library. When a search fails, some people scroll, leave or switch to another service.

AI GENIE TURNS WHAT GOOGLE ALREADY KNOWS ABOUT EACH PHOTO INTO QUESTIONS THAT FIX A VAGUE SEARCH

1 What makes it possible: a one-time scan at backup

When a photo is backed up, Gemini reads it once and saves short notes: the setting, the activity, who is in the group, clothing colours, a guess at the occasion and the time of day. Old photos are scanned once in the background. Helping you while you type is then a quick look-up of these notes, so it is fast and cheap. Both ideas use the same notes.

2 Two ideas that help while you type

Idea A: AI Genie (built)

Built

- What you see:** You type "pool". After a short pause, if the search is vague, a small box appears under the search bar with up to 3 easy questions: Who? Look? What? You can ignore it or close it.
- How it works:** It reads the saved notes of photos that match your words. It asks about the details that split those photos best, skips what you already typed, and never offers an option with zero photos. Each tap adds words to the search bar, and you press Search as usual.
- Why it helps:** You answer only what you remember, and each answer adds a detail that was missing.

Idea B: Scene Suggestions

- What you see:** As you type "pool", the search bar offers 3 to 4 full scenes, like "Friends in red swimsuits at the pool". You tap one.
- How it works:** It takes the saved scene sentences of matching photos, groups similar ones, and shows the few that cover the most different photos. Your tap replaces your short search with the full scene.
- Why it helps:** One tap adds many details. But it helps only if your memory matches a whole scene, and private photos need a filter.

3 How we chose

Both ideas pass the first test: they help while you type, so they can change what the search carries. Then we rated them on six more criteria.

Criterion	AI Genie	Scene Suggestions	Better
Adds details to the search (raises Query Formation Rate)	Medium-High (one detail per tap)	High (several per tap)	Scene Suggestions
Works when you remember only some details	High	Medium (needs a whole scene to match)	AI Genie
Easy for the user	Medium (a few taps)	High (one tap)	Scene Suggestions
Simple to build	Medium-High	Medium	AI Genie
Protects private photos	Medium-High (short option words)	Medium (full sentences need a filter)	AI Genie

4 Our choice

We chose AI Genie because it works when you remember only some details, is simpler to build, and is ready to test. Scene Suggestions needs only one tap and adds more details at once. If testers skip our questions, we will rethink.